



---

## CE 305P: Environment and Earth Science Lab

---

|               |                                    |
|---------------|------------------------------------|
| Course Code   | CE 305P                            |
| Course Name   | Environment and Earth Science Lab  |
| L-T-P-C       | 0-0-2-1                            |
| Prerequisites | None                               |
| Intended for  | UG                                 |
| Distribution  | Discipline Core                    |
| Approval      | 9th Senate (May 2016)              |
| Recoded to    | CE 306P by the 52nd BoA (Nov 2023) |

---

### Course Modules

#### Environment

- Determination of B.O.D. & C.O.D. of Wastewater Sample
- Analysis of major cations and anions
- Jar Test for Determining Optimum Coagulant Dosage
- Tests for Coliforms, Fecal, Total Coliforms
- Analysis of Metal and heavy metal in water and waste water
- Soil contamination studies
- Air Quality: Ambient Air quality (High Volume Sampler), PM10, PM2.5

#### Earth Science

- Rock and mineral identification,
- Dip/ Strike, fold, fault, fracture measurements in rocks
- Strength characterization of rocks
- Chemical analysis of rocks and soils
- Field visit

#### Text Books

1. Sawyer, N.C. and McCarty, P.L.,, **Chemistry for Environmental Engineering**, 5th Edition, McGraw Hill, 2003.
2. APHA, **Standard Methods Examination of Water and Wastewater**, American Public Health Association, 1998.